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System and method for coupling pipes

Abstract

A system for coupling pipe segments endwise to each other comprise a tubular hub secured proximate to an end face of each pipe segment, each hub having a radially extending mounting block therearound, a seal extending around the hubs between the mounting blocks and a clamping ring sized to extend around the tubular hub, said clamping ring having a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to US provisional patent application entitled “System and Method for Coupling Pipes” filed on Jul. 16, 2020 as application No. 63/052,874 and US provisional patent application entitled “System and Method for Coupling Pipes” filed on Feb. 9, 2021 as application No. 63/147,706.

BACKGROUND

1. Technical Field

(1) This disclosure relates generally to pipes and in particular to a method and system for coupling pipes.

2. Description of Related Art

(2) In conventional pipeline construction, the pipeline is formed by endwise connecting a plurality

of pipe segments together to form a continuous conduit. Such pipe segments are commonly connected through a variety of means including welding or using couplers. It will be appreciated that welding pipe segments requires a significant level of welder skill to ensure a strong liquid impervious seal between the pipe segments.

(3) Some attempts have been made to provide couplers for connecting pipe segments together. However such previous solutions have been difficult to locate and seal against the ends of the pipe segments. In particular, such previous couplers have relied upon a clamping segments secure around each end of the pipes with a sealing element thereunder. However some of such systems have been designed to engage a rings around pipe ends to hold the ends of the pipes together. Such designs have disadvantageously however hidden the surfaces to be mated for coupling the pipes together within the interior of the coupling thereby rendering connection difficult. Examples of such systems may be found in U.S. Pat. No. 8,136,847 to Madara et al.

(4) Other current designs have provided coupling segments with lips to engage within grooves of the pipes. Disadvantageously, such designs have relied on containing the seal within a cavity in the clamping ring such that the clamping segments compress the seal radially inward upon the pipe. It will be appreciated that such location of the seal within the clamping ring increases the possibility of the seal moving longitudinally out of position and also increases the possibility of a portion of the seal being damaged when compressed between the clamping segments and the pipe. Examples of such designs may be found in U.S. Pat. No. 8,500,174 to Gibb.

SUMMARY OF THE DISCLOSURE

(5) According to a first embodiment, there is disclosed a system for coupling pipe segments endwise to each other comprising a tubular hub secured proximate to an end face of each pipe segment, each hub having a radially extending mounting block therearound, a seal extending around the hubs between the mounting blocks and a clamping ring sized to extend around the tubular hub, said clamping ring having a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks.

(6) The mounting blocks extends between inner and outer annular end surfaces. The inner end surface have an inclined surface angularly oriented towards the pipe segment. The outer end surface include a transition surface at an outer radial corner thereof. The transition surface are selected from the group consisting of chamfered and radiused.

(7) The hub further includes a positioning wall extending radially therefrom parallel to and spaced apart from the mounting block. The positioning lips on the clamping ring are positioned to be received in an annular groove between the positioning wall and the outer end surface of the mounting blocks.

(8) The positioning lip may extend between inner and outer annular surfaces. The outer surfaces may be inclined relative to perpendicular to the pipe axis so as to guide the positioning lips into the annular grooves.

(9) The clamping ring may be formed of a pair of connectable semi-circular segments. The semi-circular segments are connectable to each other by fasteners.

(10) The seal may extend between first and second end surfaces. The first and second end surfaces may include an inclined portion corresponding to the inner surfaces of the mounting blocks. The first and second end surfaces may include an annular portion outward of the inclined portion. The clamping ring includes an inwardly facing surface adapted to span the mounting blocks and compress the seal between the mounting blocks.

(11) According to a further embodiment, there is disclosed a method of coupling pipe segments endwise to each other comprising securing a tubular hub proximate to an end face of each pipe segment, each hub having a radially extending mounting block therearound, locating a seal around the hubs between the mounting blocks and compressibly securing a clamping ring around the tubular hub, wherein the clamping ring have a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks.

(12) Other aspects and features of the present disclosure will become apparent to those ordinarily skilled in the art upon review of the following description of specific embodiments in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings constitute part of the disclosure. Each drawing illustrates exemplary aspects wherein similar characters of reference denote corresponding parts in each view,
- (2) FIG. 1 is a perspective view of an apparatus for coupling two pipe portions together according to a first embodiment.
- (3) FIG. 2 is a cross sectional view of the apparatus and pipe portions of FIG. 1 as taken along the line 2-2.
- (4) FIG. 3 is a detailed cross sectional view of one of the hubs of the apparatus of FIG. 1.
- (5) FIG. 4 is a cross sectional view of the clamping ring of the apparatus of Figure.
- (6) FIG. 5 is a cross sectional view of the seal of the apparatus of Figure.
- (7) FIG. 6 is a cross sectional view of the seal located between two hubs of the apparatus of FIG. 1 located around the ends of two pipes to be coupled together.
- (8) FIG. 7 is a cross sectional view of the pipes of FIG. 6 with hubs and seal located therearound and a clamping ring positioned thereover at an initial position of coupling.
- (9) FIG. 8 is a cross sectional view of the apparatus of FIG. 7 at a final position of coupling.
- (10) FIG. 9 is a detailed cross sectional view of the positioning wall of the clamping ring located within the annular groove of the hub.
- (11) FIG. 10 is a detailed perspective view of the coupling blocks of the clamping ring segments according to a further embodiment.

DETAILED DESCRIPTION

- (12) Aspects of the present disclosure are now described with reference to exemplary apparatuses, methods and systems. Referring to FIG. 1, an exemplary apparatus for coupling pipes according to a first embodiment is shown generally at 10. In particular, as illustrated in FIG. 1, the apparatus 10 couples a pair of pipe segments 8 to each other in a sealed manner so as to form a continuous pipeline therewith. It will be appreciated that a plurality of pipe segments 8 and apparatuses 10 will be utilized to form a pipeline of a desired length. It will be furthermore appreciated that the diameters of the pipe segments 8 and apparatus 10 may be selected to provide the pipeline capacity as is required for that particular pipeline and that therefore any diameter of pipe segment and apparatus 10 may be selected to fulfil those needs.
- (13) The apparatus 10 comprises a pair of hubs 12 secured to adjacent ends of the pipe segments 8 which are then coupled together by a clamping ring 60 to sealably secure the pipe segments 8. The apparatus further includes a seal ring 80 between the clamping ring 60 and the hubs 12 so as to provide a fluid impervious connection between the pipe segments 8.
- (14) Turning now to FIG. 3, each hub 12 comprises a cylindrical body extending along an axis 18 of the pipe segments between inside and outside edges, 14 and 16, respectively. The hub 12 includes an inner surface 18 sized to correspond to an outer diameter of the pipe segment so as to be closely and slidably received thereon. Once located on the pipe segment 8 with the inside edge 14 proximate to an end of the pipe segment, the hub 12 is welded or otherwise secured to the pipe segment. It will be appreciated that the hub 12 may be located proximate to an end surface of the pipe segment 8 or spaced back therefrom by a particular distance so as to secure the pipe segments together with a desired space between the end surfaces once the clamping ring 60 is secured thereover as will be set out further below.
- (15) The hub 12 includes a mounting block 30 extending radially therearound. The mounting block

30 extends between inner and outer annular surfaces, **32** and **34**, respectively. The inner surface **32** is oriented towards the inside edge **14** with a cylindrical seal supporting surface **36** therebetween. The seal supporting surface **36** may be substantially cylindrical and parallel to the axis of the pipes. Optionally the seal supporting surface **36** may include a texture such as spiralled or roughened to assist with engagement of the seal **80**. The inner surface **32** is angled relative to the axis **18** by an incline angle generally indicated at **39** selected to be between 40 and 80 degrees although other angles may be useful as well. In such a manner the inner surface forms an overhang region generally indicated at **37** located radially between the seal supporting surface **36** and the inner surface **32** which contains and holds the seal **80** in the desired position as will be more fully described below. The outer surface **34** is substantially orthogonal to the axis **18**. The top and bottom edges, **38** and **40**, respectively, of the inner surface **32** may be chamfered or radiused so as to correspond to the shape of the seal located therein and reduce stress or compression locations. Additionally, the top edge **42** of the outer surface **34** may be profiled to assist with guiding lips on the clamping ring **60** therepast. The mounting blocks **30** include an outer radial surface **31** extending between the inner surface **32** and outer surface **34**. The outer radial surface **31** is substantially cylindrical and parallel to the axis **18**.

(16) The seal supporting surface **36** in combination with the radiused bottom edge **40** and inclined inner surface **32** as illustrated in FIG. 3 may be particularly suited to receiving a seal **80** in a snug fit between a pair of opposed hubs **12** on opposed pipe segments as will be described more fully below. In particular, the profiled surface may be chamfered or radiused as are commonly known. As such, a visual representation will be provided to a worker as to whether the proper longitudinal spacing between the hubs **12** has been achieved before the clamping ring **60** is secured thereover. It will be appreciated that ensuring proper longitudinal spacing between the hubs **12** prior to attaching the clamping ring **60** will ensure that the seal **80** is fully contained between the mounting blocks **30** so as to prevent a portion of the seal from being pinched between the clamping ring **60** and the hubs **12** as the clamping ring **60** is secured thereover.

(17) As illustrated in FIG. 3, the hub **12** may also include a positioning wall **50** extending therefrom. The positioning wall **50** includes a guide surface **54** which is parallel to and spaced apart from the outer surface **34** of the mounting block **30** so as to form a circumferential groove **52** therebetween. In particular the guide surface is substantially orthogonal to the axis **18** and spaced apart from the outer surface **34** of the mounting block **30** such that the groove **52** is operable to receive a positioning lip **68** of the clamping ring **60** (as described below) in surface engagement with the outside surface **34**. In such an position the guide surface **54** will be spaced apart from the positioning lip **68** by a gap distance generally indicated at **7** as illustrated in in FIG. 9. In operation the groove **52** may be formed to have a width between the outer surface **34** and the guide surface **54** sufficient to provide a gap distance of between 0.090 and 0.1000 inches (2.2 and 2.8 mm). This gap distance **7** prevents the positioning lip **68** (as described below) from being frictionally compressed within the groove while still guiding the positioning lip **68** into the groove **52**.

(18) Turning now to FIG. 4, a cross sectional view one side of the clamping ring **60** is illustrated. The clamping ring **60** comprises a cylindrical body **62** extending between first and second ends, **64** and **66**, respectively. The clamping ring **60** includes a positioning lip **68** adjacent to each of the first and second ends **64** and **66**. Each positioning lip **68** extends radially inward from the cylindrical body between outside **70** and inside **72**, surfaces. The outside surface **70** may be angled by an angle generally indicated at **74** relative to the axis so as to be operable to engage against the guide surface **54** of the positioning wall **50** and guide the positioning lip **68** into the groove **52**. In practice an angle **74** selected to be between 12.5 and 20 degrees has been found to be useful although it will be appreciated that other angles may be useful as well. It will also be appreciated that other angles outside this range may also be useful including orthogonal to the axis **18**. The inside surface **72** is substantially orthogonal to the axis **18** so as to engage against the outer surface **34** of the positioning block **30**. Furthermore the bottom edges of each of the inside and outside surfaces **72**

and **76** may be camfered or radiused to assist with introduction of the positioning lip **68** into the groove **52**. As illustrated in FIG. 3, the positioning lips **68** and cylindrical body form an annular cavity **78** therebetween which will contain and compress the seal **80** when the clamping ring **60** is secured thereover. The clamping ring **60** includes an inner radial surface **79** extending between the positioning lips **68** also defining the annular cavity. The inner radial surface **79** is substantially cylindrical and parallel to the axis **18** and aligned with the outer radial surface **31** of the mounting blocks **30** when connected thereto. In particular as illustrated in FIG. 9, the inner radial surface **79** may engage upon the outer radial surface of the mounting blocks or substantially close thereto. As illustrated in FIG. 1, the clamping ring **60** may be formed of one or more ring segments **76** fastened together with bolts **77** or the like. The clamping rings **60** may include coupling blocks **100** at the ends thereof through which the bolts **77** are passed. As illustrated in FIG. 10, the coupling blocks **100** include at least one bolt hole **102** therethrough sized to pass a bolt of a suitable size for coupling the ring segments together. The coupling blocks **100** may include a ledge or step **104** formed therein at a position offset from the bolt holes **102**. The ledge **104** includes a retaining face **106** extending substantially parallel to an axis of the bolt holes. The retaining face **106** is spaced away from the bolt holes **102** by a distance selected to engage upon one face of a bolt **77** passed through the bolt holes **102** so as to prevent rotation thereof. It will be appreciated that this distance may be selected to permit the use of standard bolts.

(19) Turning now to FIG. 5, the seal comprises a ring-shaped member extending between end edges **81** and an inner **87** and outer **86** surface. The edge surfaces **81** are formed of a first inner portion **82** which may be angled relative to the axis **18** so as to correspond to the angle **39** of the inner surface **32** of the mounting blocks. An outer portion **84** of the end surfaces **81** are substantially orthogonal to the axis **18**. The inner surface **87** is sized to correspond to the diameter of the seal supporting surface **36** so as to be closely located therearound. The length of the seal **80** is selected to correspond to the desired distance between the inner surfaces **32** with the desired gap between the ends of the pipe segments such that the first inner portion **82** is retained or captured under the overhang of the inner surfaces **32**. The seal **80** includes a pair of grooves **88** proximate to each end having a first radial portion **90** and a second longitudinal portion **92** forming a pair of inwardly oriented rings **96** which extend towards each other. The grooves **88** also form a central lip **94** extending radially inwardly therefrom.

(20) In operation, as set out above, the hubs **12** are secured to the ends of the pipe segments **8** and welded or otherwise secured thereon. The hubs **12** are secured at the ends **9** of the pipes **8** as illustrated in FIG. 6. In particular, the hubs **12** may be spaced back from the ends **9** of the pipes **8** so as to maintain a desired pipe gap **7** between the ends **9** of the pipe. By way of non-limiting example, the distance between the outer surfaces **34** should be selected such that when the seal **80** is located between the inner surfaces **32**, the pipe gap **7** between the pipe ends **9** is maintained below a predetermined distance due to the contact of the inside surfaces **72** of the clamping ring **60** bearing upon the outer surfaces **34** of the mounting blocks **30**. By way of non-limiting example, the pipe gap **7** maybe limited to up to 0.15 inches (4 mm). Furthermore, the positioning wall **50** and in particular the guide surface **54** and the outer surface **34** are spaced apart from a first end **14** of the hub **12** so as to maintain the pipe gap **7** below a desired distance when the position lips **68** are located within the groove **52** with the inside surfaces **72** of the positioning lips **68** are in engagement upon the outer surface **34** of the mounting blocks **30**.

(21) Thereafter, the seal **80** may be located over each hub **12** so as to be positioned around the seal supporting surface **36** as the pipe segments **8** are first brought proximate to each other in their desired positions as illustrated in FIG. 6. In particular, the seal **80** is located between the inner surfaces **32** in a snug fit so as to provide a visual indication to a worker if the proper spacing has been achieved. In such a position, the seal **80** will be located between and in contact with the inner surfaces **32**. The clamping ring **60** may then be positioned around the hubs **12** and seal **80** as illustrated in FIG. 7 such that the cavity **78** contains the seal and the positioning lips **68** are guided

into the grooves **52** of the hubs.

(22) It will be appreciated that during some installations, the pipes **8** may be initially located farther apart than desired such that the gap **7** is greater than the required distance. Such spacing may be observed by a corresponding gap between the seal **80** and the inner surfaces **32** of the mounting block by a worker. The pipes **8** may then be moved closer together as before application of the clamping ring **60**. It will be appreciated that the profiled edge **42** on the mounting block **30** as well as a corresponding radiused or chamfered edge on the inner corner of the positioning lip **68** will assist to guiding positioning lip **68** into the groove **52** with reduced damage thereto such that the inside surface **72** bears on the contact with the outer surface **34** to achieve the desired gap **7** distance.

(23) As the clamping ring **60** is tightened around the hubs **12** and seal, the clamping ring bear against the outer surface **86** of the seal **80** so as to compress the seal. While being compressed, the grooves **88** are compressed and narrowed and the central lip **94** will be pressed against the ends **9** of the pipes such that a portion may be extruded or pressed into the gap **7** to form a seal between the end surfaces of the pipe segments.

(24) While specific embodiments have been described and illustrated, such embodiments should be considered illustrative only and not as limiting the disclosure as construed in accordance with the accompanying claims.

Claims

1. A system for coupling pipe segments endwise to each other comprising: a tubular hub securable annularly around and proximate to an end face of each pipe segment, each hub extending between first and second ends and having a radially extending mounting block therearound, each hub includes an integrally formed seal supporting surface located between the mounting block and the first end of the tubular hub; a seal positionable around each seal supporting surface between the mounting blocks; and a clamping ring sized to extend around the tubular hub, said clamping ring having a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks, wherein the hub and seal surround the pipe at a common longitudinal position therealong.
2. The system of claim 1 wherein the mounting blocks extends between inner and outer annular end surfaces.
3. The system of claim 2 wherein the inner end surface have an inclined surface angularly oriented towards the pipe segment.
4. The system of claim 3 wherein the outer end surface include a transition surface at an outer radial corner thereof.
5. The system of claim 4 wherein the transition surface are selected from the group consisting of chamfered and radiused.
6. The system of claim 1 wherein the hub further includes a positioning wall extending radially therefrom parallel to and spaced apart from the mounting block.
7. The system of claim 6 wherein the positioning lips on the clamping ring are positioned to be received in an annular groove between the positioning wall and the outer end surface of the mounting blocks.
8. The system of claim 7 wherein the positioning lip extends between inner and outer annular surfaces.
9. The system of claim 8 wherein the outer surfaces is inclined relative to perpendicular to the pipe axis so as to guide the positioning lips into the annular grooves.
10. The system of claim 1 wherein the clamping ring is formed of a pair of connectable semi-circular segments.
11. The system of claim 10 wherein the semi-circular segments are connectable to each other by

fasteners.

12. The system of claim 1 wherein the seal extends between first and second edge surfaces.

13. The system of claim 12 wherein the first and second edge surfaces include an inclined portion corresponding to the inner surfaces of the mounting blocks.

14. The system of claim 13 wherein the first and second edge surfaces include an annular portion outward of the inclined portion.

15. The system of claim 1 wherein the clamping ring includes an inwardly facing surface adapted to span the mounting blocks and compress the seal between the mounting blocks.

16. A method of coupling pipe segments endwise to each other comprising: securing a tubular hub proximate to an end face of each pipe segment, each hub extending between first and second ends and having a radially extending mounting block therearound, each hub includes an integrally formed seal supporting surface located between the mounting block and the first end of the of tubular hub; positioning a seal around each seal supporting surface between the mounting blocks; and compressibly securing a clamping ring around the tubular hub, wherein the clamping ring have a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks, wherein the hub and seal surround the pipe at a common longitudinal position therealong.

17. A system for coupling pipe segments endwise to each other comprising: a tubular hub securable annularly around and proximate to an end face of each pipe segment, each hub having a radially extending mounting block therearound; a seal positionable around the hubs between the mounting blocks; and a clamping ring sized to extend around the tubular hub, said clamping ring having a unitarily formed and radially extending positioning lip at each end of wherein the positioning lips surround both of the mounting blocks of the tubular hubs secured to opposed pipe segments, wherein the hub and seal surround the pipe at a common longitudinal position therealong.

18. A system for coupling pipe segments endwise to each other comprising: a tubular hub securable annularly around and proximate to an end face of each pipe segment, each hub having a radially extending mounting block therearound; a clamping ring sized to extend around the tubular hub, said clamping ring having a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks; and a seal positionable around the hubs between the mounting blocks, the seal comprising an outer cylindrical portion adapted to engage on the clamping ring, inner cylindrical portions engagable on a cylindrical portion of the tubular hubs and end portions engagable on the mounting blocks so as to form an angled path from the interior thereof, wherein the hub and seal surround the pipe at a common longitudinal position therealong.

19. A system for coupling pipe segments endwise to each other comprising: a tubular hub securable annularly around and proximate to an end face of each pipe segment, each hub having a radially extending mounting block therearound, each hub extending to an end face alignable with an end surface of each pipe segment; a seal positionable around the hubs between the mounting blocks; and a clamping ring sized to extend around the tubular hub, said clamping ring having a radially extending positioning lip at each end of wherein the positioning lips surround the mounting blocks, wherein the hub and seal surround the pipe at a common longitudinal position therealong.
